Tutorial Problem Set VII

MATA30 – TUT0003 Olle Renérius Rehnquist
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In this tutorial, we covered the chain rule, derivatives of inverse functions and implicit differentiation. Furthermore, Quiz 3 was written.

1 Basic problems

Problem 1. Compute the derivatives of the following functions.

(a) $f(x) = e^{\sqrt{3x^2+1}}$,

(b) $f(x) = x^{\arctan(x)}$,

(c) $f(x) = \sinh^{-1}(x)$ at $x = 3/4$.

Problem 2. A 5 meter long ladder leaned against a wall is sliding down vertically at a constant speed of $\frac{1}{2}$ m/s. Show that when the base of the ladder has moved 3 meters from the wall, it is sliding with speed $\frac{2}{3}$ m/s.

Problem 3. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and $g : \mathbb{R} \to \mathbb{R}$ be continuous. Later in the course, we will prove that there exists a function $G : \mathbb{R} \to \mathbb{R}$ such that $G' = g$. For now, you may assume this fact.

(a) Suppose that $y : \mathbb{R} \to \mathbb{R}$ is a solution to the differential equation

$$y'(x) = f(y(x)).$$

Using this, construct an explicit solution to the differential equation

$$z'(x) = g(x) \cdot f(z(x)).$$

(b) Find a nonzero solution to the differential equation

$$y'(x) = \frac{y(x)}{1+x^2}.$$

2 Challenging problems

Problem 4 (A singular cubic)**.** Consider the curve $C : y^2 = x^3 + x^2$. The goal of this problem is to find some nontrivial rational points on this curve.

(a) Suppose that (a, b) is a point on C. Show that the tangent line to the curve at (a, b) is given by

$$y = \left(\frac{3a^2 + 2a}{2b}\right)(x - a) + b.$$

(b) Show that if $b \neq 0$, then the tangent line to C at (a, b) intersects the curve C again in the point

$$\left(\frac{a^2}{4a+4}, -\frac{a^3(a+2)}{8b(a+1)}\right).$$

(c) Conclude that if (a, b) is a rational solution to the equation $y^2 = x^3 + x^2$ with $b \neq 0$, then $\left(\frac{a^2}{4a+4}, -\frac{a^3(a+2)}{8b(a+1)}\right)$ is also a rational solution. Could you have concluded this without any computations?

(d) Starting with the rational solution $(3, 6)$, show from the formula in (b) that the following are also rational solutions:

$$(9/16, -45/64) \quad \text{and} \quad (81/1600, 3321/64000).$$

You may use a calculator for this part.

(e) Do you think that the equation $y^2 = x^3 + x^2$ has infinitely many distinct rational points? Can you prove your answer? *Hint:* What happens to the x-coordinate if you keep iterating the formula in (b) starting with the point $(3, 6)$?

Problem 5 (Inverse function theorem)**.** Recall the statement of the inverse function theorem, as covered in lecture.

(a) Give an example of an invertible differentiable function $f : \mathbb{R} \to \mathbb{R}$ such that f^{-1} is not differentiable. Explain why this does not contradict the inverse function theorem.

(b) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is differentiable and that its derivative $f' : \mathbb{R} \to \mathbb{R}$ is continuous. Prove that if $f'(x) \neq 0$ for all $x \in \mathbb{R}$, then it follows that f^{-1} exists and is differentiable. Note that this is a stronger statement than what was given in class.

(c) Does the same conclusions hold in (b) if we do not require that f' is continuous? *Hint:* Consider the function $f(x) = x^2 \sin\left(\frac{1}{x}\right) + 2x$, extended to be 0 at the origin.

(d) **Bonus:** Learn multivariable calculus from Spivak's *Calculus on manifolds*. Deduce that the inverse function theorem is much more complicated in higher dimensions.